

PROJECT PRAHARI (■■■■■■■■■)

AI Satellite Collision Assessment & Autonomous Risk Triage Engine

Complete Codebase Architecture, Base Paper Review, Research Gaps, Feature Importance, Sample CDMs & Viva Guide

1. Codebase Architecture & How It Works (Simple & Short)

The project is structured into a clean decoupled architecture connecting a high-speed Python/FastAPI backend with a React 18 / Three.js 3D frontend:

File / Module	Type / Tech	What It Does in Simple Words
START.bat	Batch Script	1-Click Launcher: Checks Python & Node.js, auto-installs missing dependencies, launches FastAPI backend on Port 8000 and Vite frontend on Port 5173, and opens the browser.
backend/main.py	FastAPI Server	Exposes REST APIs: /api/events (archive), /api/events/{id} (details), /api/predict-cdm (custom CSV prediction), and /api/model-info (model metrics).
backend/services/prediction_service.py	ML Inference Engine	Loads trained xgboost_raw_model.pkl, aligns 100 telemetry columns, fills missing NaNs with precomputed medians, executes <2ms risk inference, and returns SHAP drivers.
backend/services/dataset_service.py	Data Service	Loads pre-indexed 2,167 historical events from data/events_summary.json, handles standalone zero-database offline fallback, and builds CDM playback timelines.
frontend/src/pages/OverviewPage.tsx	React Component	Mission control dashboard displaying live event counters (Total, Critical, High, Elevated, Low), risk distribution charts, and quick CDM file dropzone.
frontend/src/pages/EventDetailPage.tsx	Three.js / React	Interactive 3D orbital encounter scene showing Target vs Chaser trajectories, 3D covariance uncertainty bubbles, time-to-TCA slider, and parameter inspector.
frontend/src/pages/PredictionPage.tsx	React Component	Allows operators to test Individual and Sequential sample CDMs with 1-click or drag-and-drop custom CSVs to receive instant risk scores, alert bands, and top SHAP drivers.
frontend/src/pages/ModelLabPage.tsx	React Component	Interactive model playground with dynamic threshold tuning slider, confusion matrix metrics (Recall, Precision, F2), and SHAP feature importance charts.
frontend/src/pages/GlobePage.tsx	Three.js / Leaflet	3D interactive Earth globe displaying real satellite orbits (ISS, Tiangong, Sentinel) and 2D ground track trajectory overlays.

2. The Real-World Space Debris Problem

The Crisis: Over 36,000 trackable orbital debris pieces (>10 cm) travel at hypervelocity speeds exceeding **7.8 km/s (28,000 km/h)** in Low Earth Orbit (LEO). Tracking networks issue over **100,000 Conjunction Data Messages (CDMs) daily**. Over 99.9% are harmless false alarms, but human flight dynamics teams take days to manually analyze them. If a real collision threat is not identified 24 to 72 hours before closest approach (TCA), operators cannot plan and execute thruster avoidance burns.

3. What Is Mentioned in the Base Paper?

The Foundation: The project builds upon the **ESA Kelvins Collision Avoidance Challenge** conducted by the European Space Agency (ESA). The base paper investigated predicting the final collision risk ($\log_{10} P_c$) at the moment of closest approach (TCA) using intermediate, early CDMs received days prior.

Key Concepts from the Base Paper:

- Dataset:** Real & simulated conjunction events from ESA operational missions (Sentinel-1/2/3, CryoSat-2, Swarm) and space debris catalogs.
- Custom Kelvins Loss Metric (\$L\$):** An asymmetric weighted mean squared error that heavily penalizes missing high-risk events:
$$L = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \cdot w_i, \quad \text{where } w_i = 10^{\max(0, y_i + 6)}$$
- Base Paper Baselines:** Traditional 2D Foster / 3D Alfriend analytical probability integrals, Linear Regression, Multi-Layer Perceptrons (MLP), and standard Random Forests.

4. Gaps Found in the Base Paper & How Prahari Fixed Them

Identified Gap / Limitation in Base Paper	Why It Was a Problem (Flaw)	How Prahari Fixed It
1. Neglected Cross-Feature Physical Interactions	Base models looked at miss distance in isolation, missing that a 500m miss with high covariance uncertainty is far deadlier than a 100m miss with tight uncertainty.	Integrated 100 Native Telemetry Parameters: Full 3D covariance determinants, velocity vectors, and Mahalanobis uncertainty scaling.
2. High False Negative Rate (Missed Collisions)	Competing ML models used default 0.5 classification cutoffs, missing 40% to 54% of true high-risk collision events (catastrophic for satellite safety).	Calibrated -6.0 Log-Risk Threshold: Achieved 92.7% Recall on critical events (165/178 critical collisions detected early).
3. Fragility with Missing Data (Telemetry Dropouts)	Real radar tracking has frequent missing parameters (NaNs). Baseline models dropped rows or used zero-imputation, distorting covariance matrices.	Domain-Specific Median Imputation & Tree NaN Routing: XGBoost routes missing values natively without corrupting physical calculations.
4. Black-Box Nature & No Operational Visualization	Base paper outputs were static numbers with zero explainability or visual verification for flight dynamics operators under high stress.	SHAP Feature Attribution & Real-time 3D Scene: Operators see exact physical drivers and interactive 3D encounter trajectories.

5. Our Feature Importance Breakdown (What Drives the AI?)

Based on SHAP (SHapley Additive exPlanations) and XGBoost gain metrics, here are the top physical parameters governing collision risk predictions:

Feature Name	Importance	Physical Meaning & Why It Matters
Mahalanobis Distance	22.4%	3D spatial miss distance scaled by the combined positional uncertainty ellipsoids. The single strongest predictor of true physical collision probability.
Closest Miss Distance (m)	18.5%	Minimum physical Euclidean distance between target and chaser at TCA.
Relative Speed (m/s)	14.2%	Encounter velocity. Dictates the duration of the encounter window and total kinetic impact energy ($E_k = 0.5 m v^2$).
Time to TCA (days)	11.8%	Time remaining until closest approach. As time decreases, tracking uncertainty shrinks and predictions become definitive.
Covariance Determinants (Det C)	16.5%	Combined determinant of Target & Chaser position covariance matrices ($\det C_t, \det C_c$), representing total 3D uncertainty volume.
Radial Uncertainty (σ_r)	10.2%	Position error along the radial direction (altitude axis). This is the most sensitive orbital dimension for collision geometry.
Solar Radio Flux ($SF_{10.7}$)	3.5%	Solar 10.7cm flux index. Directly drives thermospheric density expansion, increasing satellite drag and orbital decay uncertainty.
Geocentric Latitude	2.9%	Orbital latitude at TCA, accounting for Earth's oblateness (J_2 gravitational perturbation effect).

6. Sample CDM Files for Testing the ML (Frontend Web Bench)

Two dedicated sample CDM files are pre-loaded in the web application (Prediction Tab) for instant 1-click evaluation:

Sample File	Type & Structure	Simulated Scenario & Expected Prediction
sample_individual_cdm.csv	Type A: Individual CDM (1 Observation Row)	Simulates a single snapshot encounter (Event 2) with tight miss distance and high position covariance. Yields HIGH risk alert ($\log_{10} P_c \approx -4.66$).
sample_sequential_cdms.csv	Type B: Sequential CDMs (5 Chronological Rows)	Simulates a multi-observation tracking sequence (Event 0) tracking risk evolution from $T=-6.84d$ to $T=-2.22d$. Yields LOW risk alert ($\log_{10} P_c \approx -7.37$).

7. Operational Alert Bands & Triage Matrix

Alert Band	Log10 Risk Threshold	Real Probability	Operational Action Required
CRITICAL	≥ -4.0	≥ 1 in 10,000	Immediate Thruster Maneuver (CAM) mandatory.
HIGH	-5.0 to -4.0	1 in 10k to 1 in 100k	Task radar tracking; compute avoidance burns.
ELEVATED	-6.0 to -5.0	1 in 100k to 1 in 1M	Monitor follow-up CDMs; stand-by status.
LOW	< -6.0	< 1 in 1,000,000	Benign approach; automated archive (no burn).

8. Top 10 Viva & Interview Questions (Quick Revision Sheet)

Q1: What is TCA and why is it critical?

Answer: TCA stands for **Time of Closest Approach**. It is the exact second when two orbital objects reach their minimum distance. Collision Avoidance Maneuvers (CAM) must be executed at least 24-48 hours before TCA to ensure trajectory clearance with minimum fuel consumption.

Q2: What is a CDM (Conjunction Data Message)?

Answer: A CDM is an international standard telemetry format (CCSDS standard) issued by space surveillance networks (ESA/SSN) containing state vectors, velocities, and 3D uncertainty covariances (σ_r , σ_t , σ_n) for two approaching objects.

Q3: Why do we express collision risk in log10 format (e.g. -4.0)?

Answer: Real collision probabilities are tiny numbers like 0.0001 (10^{-4}) or 0.000001 (10^{-6}). Expressing them as $\log_{10}(\text{Risk})$ converts these exponential fractions into an intuitive linear operational scale ($-4.0 = 10^{-4}$, $-6.0 = 10^{-6}$).

Q4: Why prioritize Recall (92.7%) over Precision?

Answer: In aerospace safety, a **False Negative (missed collision) is fatal** (satellite destruction and Kessler cascade). A False Positive only causes an engineer to verify the data. High Recall guarantees no critical threats slip through unnoticed.

Q5: Why choose XGBoost over Deep Neural Networks for this task?

Answer: CDMs are tabular telemetry with 100 heterogeneous numerical features and missing tracking data (NaNs). XGBoost natively handles missing values, trains in seconds, runs inference in $<2\text{ms}$, and outperforms deep neural networks on tabular datasets.

Q6: What is the Mahalanobis Distance?

Answer: It measures the distance between two objects **scaled by their combined 3D covariance uncertainty bubbles**. A 500m miss distance inside a 2,000m uncertainty bubble is dangerous, whereas a 500m miss with a 10m uncertainty bubble is safe.

Q7: What is the Kessler Syndrome?

Answer: A runaway collision chain reaction where orbital debris collisions create more debris fragments, permanently destroying access to Low Earth Orbit for generations.

Q8: How does Prahari handle multiple CDMs over time for a single event?

Answer: As ground radars make newer observations closer to TCA, multiple CDMs are issued. Prahari tracks the chronological risk evolution and evaluates the latest available telemetry state for the most accurate prediction.

Q9: What is the difference between Individual and Sequential CDM testing?

Answer: Individual CDMs test single-epoch alert triage (snapshot encounter), while Sequential CDMs test time-series tracking updates as uncertainty shrinks approaching TCA. Both can be tested with 1-click in the Prediction page.

Q10: What makes Prahari a standalone deployment?

Answer: Prahari packages its pre-computed 2,167 event archive and ML model weights locally. Double-clicking START.bat automatically verifies dependencies and launches both backend and frontend without external cloud databases.